

External Surface Enumeration Detection Architecture

This document explains how the **Reconnaissance External Surface Enumeration** use case works inside the Sentinel.

Detection Objective

Detect when a Windows endpoint performs **multiple reconnaissance commands** within a short time window, such as:

- nslookup.exe
- ping.exe
- tracert.exe
- netstat.exe
- arp.exe

This simulates **MITRE ATT&CK – Reconnaissance/Active Scanning**.

High-Level Architecture



Component-by-Component Flow

1) Endpoint Attack Simulation

The user runs safe recon commands:

```
nslookup google.com
ping 8.8.8.8
tracert google.com
```

Each command launches its own executable.

2) Sysmon Telemetry Generation

Sysmon captures **New Process Created** → Event ID 1.

Important fields:

- CommandLine
- Image
- User
- ParentImage
- UtcTime

Example event:

```
Image: C:\Windows\System32\PING.EXE
CommandLine: ping 8.8.8.8
```

3) AMA + DCR Collection Layer

Azure Monitor Agent collects:

```
Microsoft-Windows-Sysmon/Operational
```

using your custom **Data Collection Rule (DCR)**.

The event is forwarded to:

```
Log Analytics Workspace → SecurityEvent
```

4) KQL Detection Logic

The analytics rule does 4 major things:

A) Filter only Sysmon Process Create

```
| where Channel == "Microsoft-Windows-Sysmon/Operational"
| where EventID == 1
```

B) Extract executable name

```
| extend ProcessName = tostring(tolower(split(CommandLine, " ")[0]))
```

C) Keep only recon tools

```
| where ProcessName has_any (
  "nslookup.exe",
  "ping.exe",
  "tracert.exe",
  "netstat.exe",
  "arp.exe"
)
```

D) Correlate repeated tools

```
| summarize ReconCount = count() by Computer
| where ReconCount >= 3
```

This prevents single-command noise.

Alert Trigger Logic

The alert fires only when:

3 or more recon commands are executed within 15 minutes

This improves fidelity and reduces false positives.

Examples:

Alert

nslookup + ping + tracert

No Alert

single ping.exe

Final SOC Workflow

```
Attack Simulation
↓
Sysmon Event 1
↓
AMA + DCR
↓
SecurityEvent table
↓
KQL correlation
↓
Analytics Rule
↓
Sentinel Incident
↓
MITRE ATT&CK reporting
```

This is the full working detection pipeline for your GitHub project.